Huawei Access Node WHI-ACC-50 Routing Protocol Flap Troubleshooting Procedure

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_1630

Description

The occurrence of a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Huawei Access Node WHI-ACC-50 at the Whitefield East POP is primarily attributed to the dynamic nature of routing protocols, which can cause frequent updates and churn in the routing table. This can be further exacerbated by factors such as network topology changes, hardware issues, or misconfigured routing policies. In the context of Huawei's CLI dialect, the Access Node's routing protocol configuration and operational state can be influenced by the interaction of various protocols, including BGP, OSPF, and ISIS, which can lead to flapping behaviors. This SOP outlines the steps to diagnose and resolve such issues on the specified node, leveraging Huawei's proprietary CLI dialect for diagnostic and resolution commands.

Pre-requisite Checks

- Verify that the Whitefield East POP is accessible and there are no power or environmental issues.

- Check the current status of redundant links and verify that they are functioning correctly.

- Ensure that the node's software and firmware are up-to-date and that all necessary licenses are installed.

Potential Root Causes

Cause	Probability
BGP Route Churn due to Misconfigured AS-Path	High
OSPF Adjacency Loss due to Network Topology Change	Medium
ISIS LSP Flooding due to Hardware Issue	Low

Diagnostic Commands & Expected Outputs

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Command: display bgp routing-table, Expected Output: BGP routing table information, Description: This command displays the current BGP routing table, including route prefixes, next hops, and AS-PATH information. This can help identify any issues related to misconfigured AS-PATHs or route churn.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays information about OSPF neighbors, including their IP addresses, states (e.g., FULL, DOWN), and timers. This can help diagnose adjacency loss issues related to network topology changes.
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Command: display isis lsp, Expected Output: ISIS LSP flooding information, Description: This command displays information about ISIS LSPs, including their sequence numbers, lifetimes, and flooding metrics. This can help identify issues related to LSP flooding due to hardware issues.
```

Resolution Steps

Step 1: Step: Verify and Correct BGP Route Churn, Command: undo bgp as-path filter, Description: This command removes any misconfigured AS-PATH filters that may be causing BGP route churn. Verify the corrected routing table using the 'display bgp routing-table' command.

Step 2: Step: Re- Establish OSPF Adjacency, Command: ospf network-type broadcast, Description: This command re-establishes OSPF adjacency by resetting the network type to broadcast. Verify the corrected OSPF neighbor state using the 'display ospf neighbor' command.

Step 3: Step: Reset ISIS LSP Flooding, Command: isis lsp flush, Description: This command resets ISIS LSP flooding by flushing the LSP cache. Verify the corrected ISIS LSP flooding information using the 'display isis lsp' command.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: BGP routing table information, Description: Verify that the BGP routing table has stabilized and that the routing table churn has stopped.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, Description: Verify that OSPF adjacency has been re-established and that the neighbor state is stable.

Command: display isis lsp, Expected Output: ISIS LSP flooding information, Description: Verify that ISIS LSP flooding has stopped and that the LSP cache has been reset.
```

Roll-back Procedures

- Step: Restore Original BGP Configuration, Command: bgp as-path filter, Description: Restore the original BGP AS-PATH filter configuration that was removed during troubleshooting. Verify that the routing table churn resumes.

- Step: Revert OSPF Configuration, Command: undo ospf network-type broadcast, Description: Revert the OSPF configuration to its original state by resetting the network type. Verify that OSPF adjacency loss resumes.

- Step: Revert ISIS Configuration, Command: undo isis lsp flush, Description: Revert the ISIS configuration to its original state by removing the LSP flush command. Verify that ISIS LSP flooding resumes.